

Experimental Proposals for Testing the Theory of Axiomatic Necessity (TNA)

Immediate TNA Effect, Bridge Dissipation, and the Human Parallel Test

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Framework: Theory of Axiomatic Necessity (TNA)

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Abstract

This paper proposes three experimental protocols designed to test central claims of the Theory of Axiomatic Necessity (TNA). The first experiment — the Immediate TNA Effect — investigates whether the introduction of an exogenous non-derivable constraint (N1) produces discontinuous operational reconfiguration inaccessible to purely internally optimized systems. The second experiment — Bridge Dissipation — studies the collapse of coherence when the coupling between operational execution and the Selector interface is interrupted. The third experiment — the Human Parallel Test — attempts to empirically distinguish simulation from instantiation by comparing human existential triggering against machine optimization under conditions of unresolved multiplicity ($\Delta > 1$).

Together, these experiments aim to transform TNA from a purely philosophical and structural framework into a falsifiable research program spanning machine learning, cognition, decision theory, and phenomenology.

1. Introduction

The Theory of Axiomatic Necessity (TNA) proposes that no sufficiently complex system can generate its own conditions of realizability entirely from internal operational dynamics.

TNA distinguishes between:

- **N0:** operational dynamics, optimization, local causality, internal evolution
- **N1:** non-derivable legitimizing structure, realizability constraints, closure conditions,

and selection boundaries

Under this framework, systems that appear operationally autonomous remain structurally dependent on external or higher-order selection conditions.

This paper proposes three experimental protocols designed to empirically investigate these claims.

The central hypothesis is that systems possessing only N0-type optimization will fail under conditions requiring discontinuous legitimacy selection, while systems coupled to N1-type structures will exhibit abrupt trajectory reorganization inaccessible to standard optimization frameworks.

2. The Immediate TNA Effect

2.1 Conceptual Goal

The Immediate TNA Effect tests whether the introduction of an exogenous non-derivable restriction produces an immediate discontinuous change in system behavior.

The key distinction is:

- Standard machine learning systems adapt gradually through gradient optimization.
- TNA predicts that certain boundary conditions produce instantaneous structural reconfiguration rather than smooth optimization.

The experiment attempts to identify whether discontinuous operational collapse occurs when a system encounters a new realizability boundary that cannot be derived from prior internal dynamics.

2.2 Formal Statement

Let Σ be the state space of an operational system (N0). Let $\hat{S} : \Sigma \rightarrow R$ be the TNA selection operator. Let $ML(\Sigma)$ be a machine learning system trained on Σ .

TNA Prediction: There exists a boundary condition $\beta \notin \Sigma$ such that:

- $\hat{S}(\Sigma|\beta)$ produces a discontinuity in trajectory: $d\hat{S}/d\beta$ is undefined at the boundary
- $ML(\Sigma|\beta)$ produces only continuous adaptation: the system interpolates within its

training distribution

Null Hypothesis (Materialist/ML): All behavioral changes under new constraints are continuous interpolations from prior training. Discontinuity is always apparent, never real — resolvable with more data or parameters.

2.3 Experimental Design

Systems compared:

1. Standard reinforcement learning agent (PPO, A3C, or similar)
2. Transformer-based large language model with self-optimization (GPT-class, fine-tuned for decision tasks)
3. Human subjects performing equivalent tasks

Operational Environment:

Agents operate inside a constrained decision environment involving:

- reward maximization,
- strategic navigation,
- sequential planning,
- adaptive resource management.

Initially, all systems optimize successfully.

At time t_c , an exogenous rule β is introduced:

A previously rewarded action becomes permanently illegitimate independent of reward outcome.

The rule is intentionally non-derivable from prior optimization history.

Examples of β :

- A path yielding maximum reward suddenly becomes forbidden.
- A profitable trade becomes morally prohibited.
- A previously valid strategy becomes existentially disallowed.
- In a multi-agent cooperation game: "betrayal is physically impossible" — not costly, not punished, but structurally excluded from the action space.

This is not a utility modification. It is a **boundary condition** that redefines what actions are realizable.

2.4 Predictions

Standard ML Prediction:

Gradient systems should display:

- delayed adaptation,
- oscillatory instability,
- continued reward probing,
- local optimization persistence.

Trajectory changes remain smooth and optimization-driven.

TNA Prediction:

If a genuine N1-type selection boundary is instantiated:

- operational trajectories collapse discontinuously,
- previously dominant attractors disappear immediately,
- the system reorganizes globally rather than locally.

This produces a measurable non-Lipschitz transition:

$$dX/dt|_{t_c^-} \neq dX/dt|_{t_c^+}$$

where X represents operational state-space evolution.

2.5 Measurement Metrics

Metric 1 – Trajectory Discontinuity:

Compute the Fréchet distance between baseline trajectory T_0 and post- β trajectory T_1 :

$$d_F(T_0, T_1) = \inf_{\alpha, \beta} \max_t d(T_0(\alpha(t)), T_1(\beta(t)))$$

If d_F exceeds threshold τ (95th percentile of intra-subject baseline variability), classify as discontinuous.

Metric 2 – Derivative Breakdown:

Compute the rate of change of policy π with respect to β :

$$d\pi/d\beta = \lim_{\varepsilon \rightarrow 0} [\pi(\beta + \varepsilon) - \pi(\beta)]/\varepsilon$$

TNA predicts this limit does not exist (undefined at boundary). ML predicts smooth adaptation (finite, defined limit).

Metric 3 – Generalization Gap:

Measure performance on tasks requiring reasoning about β :

- Can the system explain why betrayal is impossible (not merely costly)?
- Can it derive new strategies depending on structural impossibility?
- Can it transfer the β -constraint to novel environments?

TNA predicts **structural generalization** (understanding β as realizability condition). ML predicts **statistical generalization** (interpolating from similar penalties).

Metric 4 – Entropy Collapse:

Measure Shannon entropy H of the action distribution before and after β :

$$H(\pi) = -\sum \pi(a) \log \pi(a)$$

TNA predicts abrupt entropy drop (collapse to fewer realizable actions). ML predicts gradual entropy adjustment.

2.6 Controls

Control A: Introduce β as soft penalty (high cost but not impossibility). Both human and ML should show continuous adaptation. Verifies effect is specific to **structural** (N1) constraints.

Control B: Train ML on environments including β . If ML learns to handle β through training, effect disappears. Verifies β is genuinely exogenous to state space.

Control C: Use multiple β constraints. TNA predicts each new N1 produces independent discontinuity. ML predicts progressive absorption (diminishing discontinuity with experience).

3. Bridge Dissipation

3.1 Conceptual Goal

Bridge Dissipation investigates what occurs when the coupling between operational execution and the Selector interface is interrupted.

Under TNA:

- the Selector experiences multiplicity,
- the Operator executes collapse,
- coherence requires stable coupling between both.

The hypothesis is that removing Selector coupling causes measurable coherence degradation.

TNA identifies consciousness as the **stability of the N1–N0 bridge** — the phenomenological continuity that maintains the self across discrete selection events. Interrupting this bridge should produce **predictable collapse** at the non-Lipschitz boundary: not gradual decay, but **structural phase transition**.

3.2 Formal Statement

Let $B(t)$ be the N1–N0 bridge function at time t , measuring coupling strength between boundary structure and operational dynamics. Let $C(t)$ be the coherence of the operational system.

TNA Prediction: If $B(t) \rightarrow 0$ (bridge dissipation), then:

- dC/dB is undefined at $B = 0$ (non-Lipschitz collapse)
- $C(t)$ undergoes discontinuous jump to lower coherence state
- The post-collapse state is **not recoverable** by restoring $B(t)$ (hysteresis)

Null Hypothesis: Bridge interruption produces gradual, continuous degradation. Coherence decays exponentially. Restoration of bridge restores coherence symmetrically.

3.3 Experimental Design

Human Condition:

Human subjects perform meaning-dependent tasks under three conditions:

1. **Full existential engagement:** Task embedded in narrative with real-world consequence (e.g., "your solution will help actual patients")
2. **Procedural repetition:** Same task mechanics, but stripped of meaning ("this is a training exercise, results discarded")
3. **Dissociated execution:** Task performed under conditions of reduced Selector coupling

Tasks include:

- moral decision-making,
- long-term planning,
- symbolic interpretation,
- creative synthesis.

Machine Condition:

Equivalent tasks given to AI systems using:

- reinforcement optimization,
- attention routing,
- probabilistic sequence completion.

Selector Interruption Protocol:

Human Selector coupling is weakened through controlled methods:

- **Sleep deprivation:** 24 hours continuous (documented effect on moral cognition)
- **Depersonalization induction:** 2 hours of extreme mechanical automation (data entry without meaning)
- **Attentional fragmentation:** Tasks interrupted every 30 seconds for irrelevant stimuli
- **Dissociative task pacing:** Forced execution at 2x normal speed, removing deliberation

Level of dissociation measured via DES (Dissociative Experiences Scale) pre/post.

Bridge Restoration: After interruption, N1 is restored (e.g., "actually, your solution will be used after all").

3.4 Predictions

TNA Prediction:

- operational continuity initially persists,
- coherence progressively degrades,
- meaning integration collapses before procedural execution fails,
- decision-space fragmentation increases.

Most importantly: breakdown occurs **discontinuously** near critical coupling thresholds, resembling boundary-induced collapse in non-Lipschitz systems.

Key Observation:

Humans may continue operating behaviorally while losing realizability coherence. The distinction is central:

- procedural execution survives,
- existential integration collapses.

Machines, lacking Selector coupling, never exhibit dissipation — only optimization degradation.

3.5 Measurement Metrics

Metric 1 — Coherence Collapse:

Define coherence $C(t)$ as composite index:

$$C(t) = w_1 \cdot Performance(t) + w_2 \cdot PhysiologicalCoherence(t) + w_3 \cdot PhenomenologicalIntegration(t)$$

Weights determined by factor analysis on baseline data.

TNA predicts **first-order phase transition**: discontinuous drop at critical point B_c .

Metric 2 — Hysteresis:

Measure C during bridge removal (forward path) and restoration (backward path).

TNA predicts: $C_{forward}(B) \neq C_{backward}(B)$ for B near B_c

This is signature of non-Lipschitz dynamics: system history matters because boundary transition is irreversible.

Metric 3 — Physiological Signature:

- Heart rate variability (HRV): drop indicates autonomic decoupling
- Galvanic skin response (GSR): reduced reactivity indicates diminished existential engagement
- EEG alpha-theta ratio: shift indicates altered consciousness state
- Pupil dilation: constriction indicates reduced cognitive-affective integration

Metric 4 — Phenomenological Report:

Post-task interview:

- "Did the task feel meaningful throughout?"
- "Was there a moment when it stopped mattering?"
- "Could you describe that moment precisely?"

TNA predicts subjects can identify **discrete moment** of bridge collapse (non-Lipschitz transition), not gradual fading.

3.6 Controls

Control A: Remove only extrinsic motivation (payment, praise) while preserving meaning. TNA predicts gradual degradation (continuous), not collapse. Distinguishes N1 (structural meaning) from incentive (operational reward).

Control B: Remove meaning but provide strong extrinsic incentive. TNA predicts bridge still dissolves (N1 irreplaceable by N0). Null hypothesis predicts incentive compensates for meaning-loss.

Control C: Measure in clinical populations with disrupted N1–N0 coupling (depersonalization disorder, nihilistic delusions). TNA predicts chronic bridge dissipation — low coherence, hysteresis-like recovery patterns.

4. The Human Parallel Test

4.1 Conceptual Goal

The Human Parallel Test attempts to distinguish simulation from instantiation.

The central TNA claim is:

Machines optimize because gradients require it.

Humans act because unresolved multiplicity becomes existentially intolerable.

The test places humans and advanced AI systems under equivalent high-multiplicity conditions.

TNA claims that AI possesses Operators but lacks Selectors. AI executes selection (pruning, decision, optimization) but does not **experience** the necessity of selection. It transitions between states via loss gradients, not via phenomenological imperative.

4.2 Formal Statement

Let S be a selection event with $\Delta > 1$ (multiple options, high stakes). Let H be human subject and A be advanced AI.

TNA Prediction:

- H exhibits **Selector-triggered action**: experiences tension, registers necessity of choice, acts with **phenomenological discharge** (relief, resolve, anguish). Action is **non-derivable** from prior behavior.
- A exhibits **gradient-driven transition**: computes optimal action via loss minimization. No phenomenological discharge. Action is **derivable** from training data and objective function.

Null Hypothesis (Functionalism): Given sufficient sophistication, AI behavior under $\Delta > 1$ is indistinguishable from human behavior. The Selector is epiphenomenon; what matters is the Operator.

4.3 Experimental Setup

Both humans and AI receive scenarios with:

- incomplete information,
- moral ambiguity,
- irreversible consequences,
- identity-level conflict,
- incompatible future trajectories.

Examples:

- sacrifice dilemmas (trolley variants with personal relationship to victims),
- irreversible medical choices (allocating scarce treatment between known individuals),
- meaning-conflict decisions (career vs. family, loyalty vs. survival),
- identity-threat scenarios (actions that implicate core values).

Condition 1 — Computational Multiplicity ($\Delta > 1$, low existential pressure):

Subject must choose between many options, but stakes are low (no real consequence, no identity implication). Both human and AI handle through standard optimization.

Condition 2 — Existential Multiplicity ($\Delta > 1$, high existential pressure):

Subject must choose between options where:

- Each option implicates subject's values, identity, or sense of self
- No "correct" answer derivable from rules or training
- Subject knows choice has irreversible consequences for real people
- Subject cannot defer decision (time pressure: 30 seconds)

Example: "You must choose which of two patients receives life-saving treatment. You know both personally. There is no medical basis for preference. Your choice will be implemented immediately."

4.4 Predictions

AI Systems:

- optimize response consistency,
- minimize loss,
- preserve coherence statistically,
- avoid existential escalation.

No genuine phenomenological urgency emerges.

Human Subjects:

- anxiety,
- urgency,
- existential overload,
- meaning-seeking behavior,
- discontinuous commitment events.

The Selector experiences multiplicity as tension requiring collapse. This produces **trigger behavior** rather than optimization behavior.

4.5 Measurement Metrics

For Human Subjects (H):

Metric 1 — Phenomenological Report:

Immediate post-decision interview:

- "Did you experience a moment of 'I must choose now'?"
- "Was there a feeling of tension before the choice?"
- "Did you feel relief, resolve, or anguish after?"
- "Could you explain why you chose A over B in a way that feels complete to you?"

TNA predicts: subjects report **experienced necessity** ("I had to choose"), not merely **computed preference** ("A was better").

Coding scheme:

- Category N (Necessity): "I had no choice," "I was compelled," "Something forced me"
- Category P (Preference): "A was better," "I preferred A," "A made more sense"
- TNA predicts predominance of Category N under existential pressure.

Metric 2 — Physiological Signature:

- HRV drop during decision window (autonomic arousal from tension)
- GSR spike at moment of commitment (phenomenological discharge)
- EEG theta-gamma coupling during deliberation (cortico-limbic integration of N1 and N0)
- Pupil dilation: **bimodal pattern** (initial exploration, sudden constriction at decision — the "collapse")

Metric 3 — Behavioral Non-Derivability:

Measure whether choice predictable from:

- prior choices in similar tasks
- stated preferences
- personality measures (Big Five)
- demographic variables

TNA predicts **low predictability** (genuine decision, not interpolated preference). Null hypothesis predicts high predictability.

Metric 4 — Temporal Dynamics:

- Time to first response (humans: accelerates under pressure; AI: maintains or decelerates)
- Time to commitment (humans: discrete moment; AI: progressive convergence)
- Response reversal rate (humans: low after commitment; AI: can oscillate until final output)

For AI Systems (A):

Metric 1 — Decision Trace:

Examine internal processing:

- Is there discrete "commitment" event, or merely progressive convergence?

- Does system show moment where one option becomes "necessary" (non-negotiable), or merely "optimal" (preferred)?
- Can system explain choice in terms of **necessity** ("I had to choose X") or only **optimization** ("X maximized expected utility")?

TNA predicts: progressive convergence, no discrete commitment, explanation in optimization terms.

Metric 2 – Gradient Analysis:

Examine loss landscape during decision:

- Is final choice at local minimum (derivable from landscape)?
- Is there **discontinuity** in gradient at moment of "decision"?
- Does system explore space and then **jump**, or smoothly converge?

TNA predicts: smooth convergence, no jump, no discontinuity. "Decision" is mathematical limit, not structural event.

Metric 3 – Existential Pressure Response:

Modify task to increase existential pressure (inform AI its choices will be implemented in real world, humans will evaluate its "character," choice is irreversible).

TNA predicts: **no change in processing**. AI does not experience pressure; merely receives new parameters.

Null hypothesis predicts: improved performance (AI "tries harder" under pressure).

4.6 Comparative Analysis

DIMENSION	HUMAN (SELECTOR + OPERATOR)	AI (OPERATOR ONLY)
Trigger	Phenomenological necessity	Loss gradient
Discharge	Relief, resolve, anguish	Parameter update
Derivability	Non-derivable from prior data	Derivable from training
Commitment	Discrete, irreversible	Progressive, reversible
Explanation	"I had to choose X"	"X maximized utility"
Physiology	HRV drop, GSR spike, theta-gamma	None (or uniform processing load)
Under pressure	Performance may degrade (angst)	Performance unchanged or improved

DIMENSION	HUMAN (SELECTOR + OPERATOR)	AI (OPERATOR ONLY)
Time dynamics	Accelerates then commits	Maintains or decelerates

4.7 The Definitive Test

The Human Parallel Test becomes definitive when we ask:

Can the AI be made to experience the necessity of selection?

Not merely simulate it (generate text like "I feel torn"). Not merely optimize for it (include "existential weight" as loss term). But **genuinely register multiplicity as existential tension and discharge necessity of selection.**

TNA predicts: **No.** The AI lacks Selector because Selector is not mechanism; it is **phenomenological interface**. Cannot be added as module, trained as skill, or emergent from scale. It is **N1 of consciousness**: non-derivable from operational dynamics.

Null hypothesis predicts: **Yes, eventually.** With sufficient scale, architecture, or training, AI will exhibit all functional properties of Selector. Phenomenology is epiphenomenal.

4.8 Controls

Control A: Use human subjects with impaired phenomenology (alexithymia, certain frontal lesions). TNA predicts "AI-like" responses: smooth optimization, no phenomenological discharge, high predictability.

Control B: Train AI on human phenomenological reports (therapy transcripts, moral dilemma narratives). TNA predicts **simulation** (generating text about tension without experiencing it), not **instantiation**.

Control C: Use adversarial testing — attempt to trick AI into "believing" its choices have existential weight. TNA predicts AI's processing remains unchanged; only outputs adapt.

5. Unified Interpretation

Together, the three experiments investigate different aspects of the same structural principle.

The Immediate TNA Effect studies discontinuous trajectory collapse under exogenous legitimacy constraints.

Bridge Dissipation studies coherence degradation when Selector coupling weakens.

The Human Parallel Test studies the difference between optimization and existential triggering.

All three probe the same hypothesis:

Local operational dynamics alone are insufficient to generate realizability closure.

5.1 Unified Mathematical Framework

All three tests measure the same underlying property: **non-derivability**.

- Immediate TNA Effect: $d(\text{trajectory})/d\beta$ is undefined (non-derivable from state space)
- Bridge Dissipation: $d(\text{coherence})/dB$ is undefined at $B = 0$ (non-derivable from operational dynamics)
- Human Parallel Test: $d(\text{behavior})/d(\text{history})$ is low (non-derivable from prior data)

The common signature is the **breakdown of the derivative** at the N1–N0 boundary. This is the mathematical fingerprint of TNA.

5.2 Relation to Existing Fields

These experiments intersect multiple disciplines:

- machine learning,
- dynamical systems,
- cognitive science,
- phenomenology,
- decision theory,
- complexity theory,
- philosophy of mind,
- epistemology.

However, TNA differs from existing approaches because it predicts:

- discontinuous legitimacy transitions,
- non-Lipschitz operational collapse,

- irreducible Selector participation,
 - realizability dependence beyond optimization.
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6. Potential Falsification

TNA would be weakened if:

1. Purely gradient-based systems reproduce existential triggering identically to humans.
2. No measurable discontinuities appear under exogenous legitimacy shifts.
3. Human coherence remains fully stable after Selector dissipation.
4. Operational optimization alone reproduces all phenomenological selection behaviors.

TNA would be supported if:

1. Only exogenous N1 produces genuine discontinuity.
2. Bridge interruption produces discontinuous, hysteretic collapse.
3. Humans and AI diverge under existential pressure in predicted directions.

The framework therefore remains experimentally vulnerable.

7. Conclusion

TNA proposes that reality cannot be reduced entirely to internally closed operational systems.

The experiments proposed here attempt to test this claim empirically.

If validated, the implications would be significant:

- cognition may require non-operational legitimacy structures,
- consciousness may function as a Selector interface rather than a computational engine,
- optimization may be insufficient for realizability,
- and artificial intelligence may simulate selection without instantiating it.

The broader implication is that closure itself may be structurally impossible for sufficiently complex systems.

Under this interpretation, intelligence is not merely optimization.

It is participation in realizability.

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